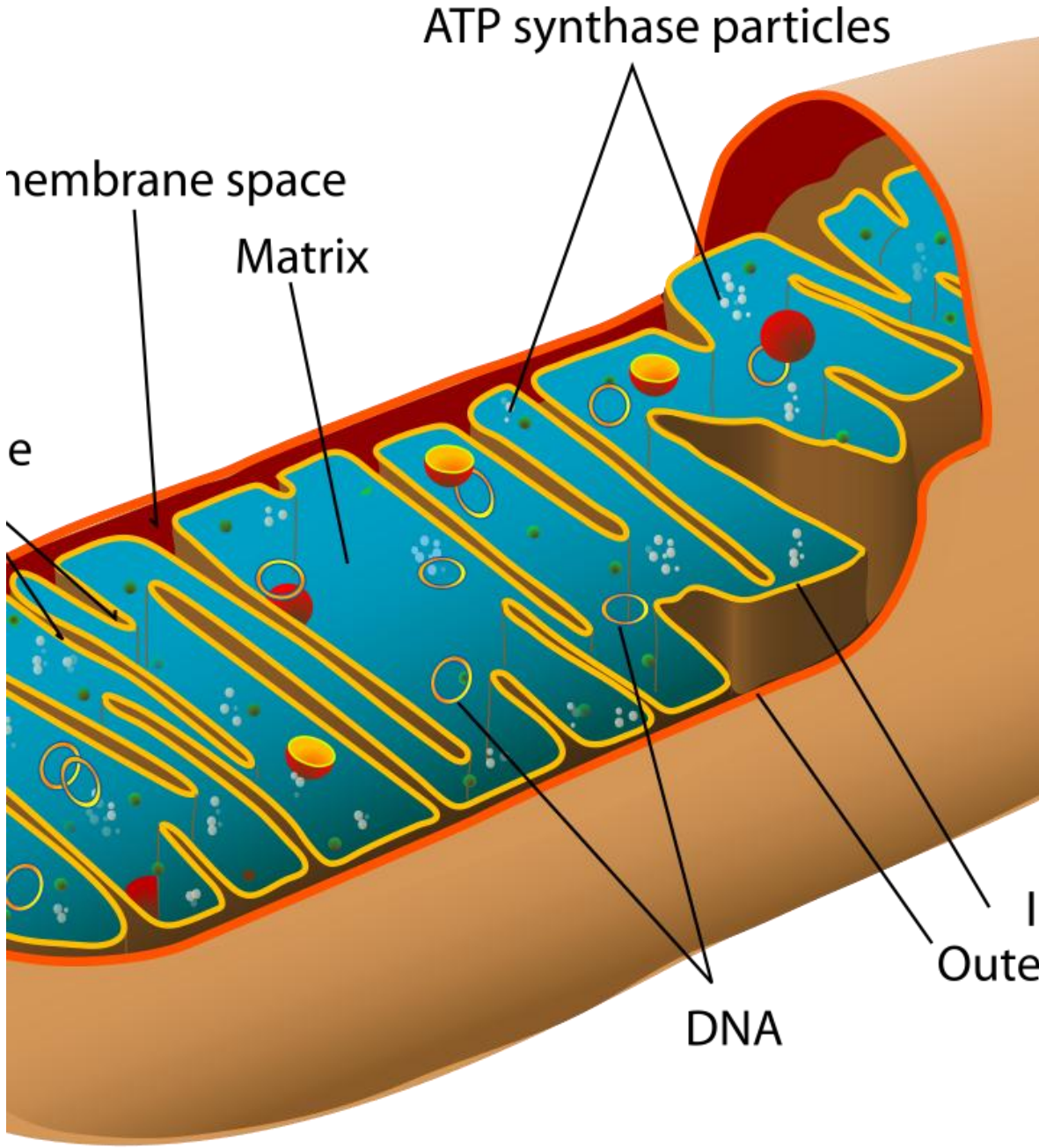
DP Biology – C1.2 (HL extension)

Inside cell respiration

- What do hydrogen and oxygen contribute to releasing energy in cells?
- How is energy shared out and used within cells?



HL statements C1.2.7–17. Textbook pp. 412–424.
Image: a mitochondrion. Mariana Ruiz Villarreal, public domain.

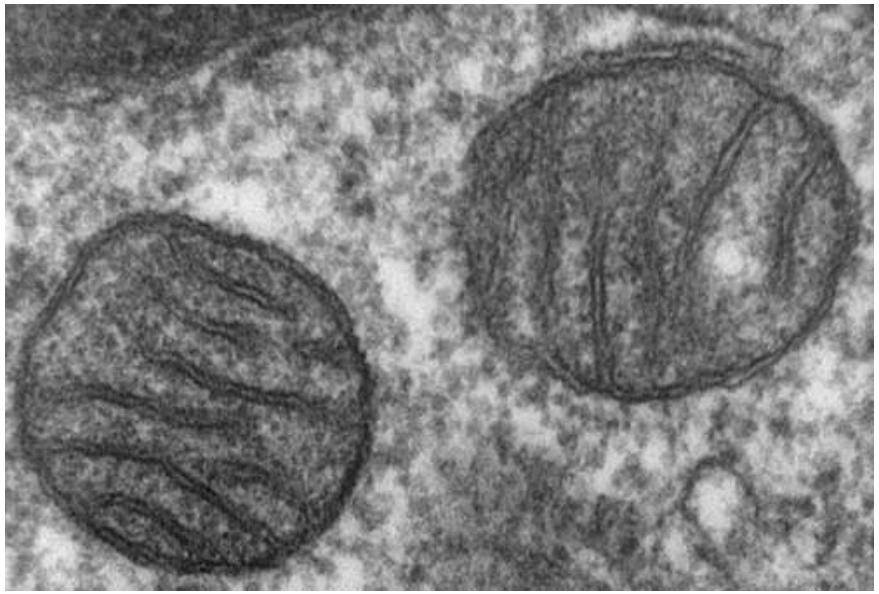
Objectives for each class

Class	Statements	By the end of the class, you can...
1	C1.2.7–10	Explain redox and NAD, glycolysis, and how anaerobic respiration regenerates NAD in humans and yeast
2	C1.2.11–14	Explain the link reaction and Krebs cycle, and how the electron transport chain builds a proton gradient
3	C1.2.15–17	Explain chemiosmosis and the role of oxygen, and compare fats and carbohydrates as fuels

The big picture

C1.2.7–16

Glycolysis
cytoplasm
→ **Link reaction**
matrix
→ **Krebs cycle**
matrix
→ **Electron transport chain and chemiosmosis**
inner membrane



Mitochondria (TEM). Louisa Howard, public domain.

Glucose is taken apart in **small, enzyme-controlled steps**, not burned in one go.

At several steps **hydrogen is removed** and loaded onto carrier molecules.

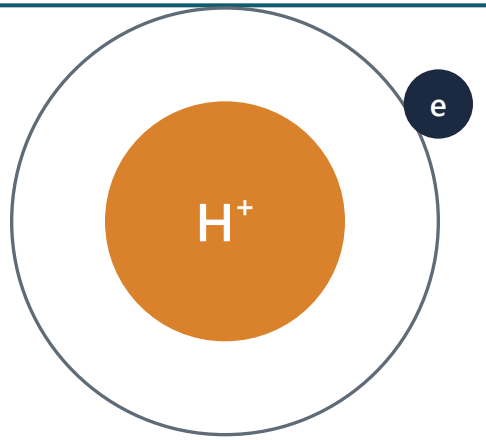
Almost all the ATP is made at the last stage, when that hydrogen meets **oxygen**.

Overall

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy. About 30–32 ATP per glucose with oxygen; only 2 without.

Oxidation = losing hydrogen

	Oxidation	Reduction
Electrons	lost	gained
Hydrogen	removed	added
Oxygen	added	removed



hydrogen atom = proton + electron

So removing hydrogen removes **electrons**: that is **oxidation**.

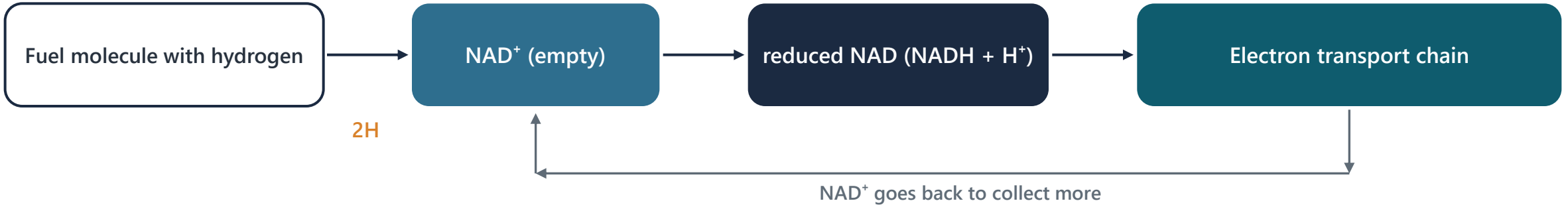
In respiration

Glucose loses its hydrogen step by step (it is oxidized). Oxygen gains it at the very end, forming water (it is reduced). Oxidation and reduction always happen together: **redox**. Enzymes that remove hydrogen are called **dehydrogenases**.

Memory aid: OIL RIG – Oxidation Is Loss, Reduction Is Gain (of electrons).

NAD and FAD: hydrogen carriers

C1.2.7



$$NAD^+ + 2H^+ + 2 \text{ electrons} \rightarrow NADH + H^+$$

NAD is a **coenzyme**: it picks up hydrogen (and its energy) from fuel molecules and delivers it to the electron transport chain, like a taxi.

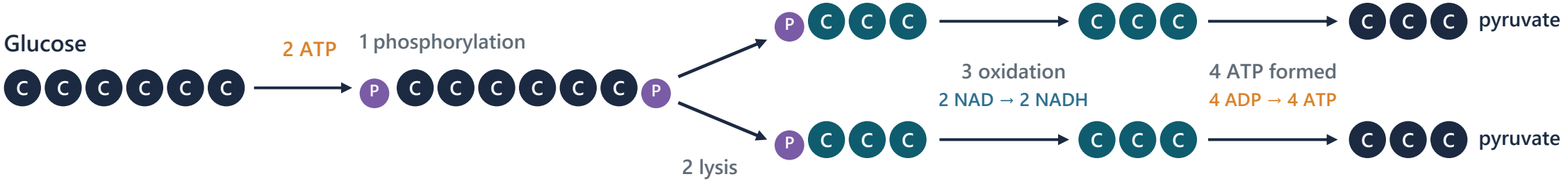
A cell has only a little NAD, so it must be **recycled** constantly. Remember this: it explains anaerobic respiration.

FAD

A second carrier, reduced to $FADH_2$ once in each turn of the Krebs cycle. It leads to less ATP than NAD (1.5 vs 2.5).

Glycolysis: glucose to two pyruvate

C1.2.8



Step	Per glucose	What happens
1 Phosphorylation	-2 ATP	Phosphates make the sugar reactive
2 Lysis	–	6C sugar splits into two 3C sugars
3 Oxidation	+2 NADH	Hydrogen removed, given to NAD^+
4 ATP formation	+4 ATP	Phosphates passed to ADP

Net result

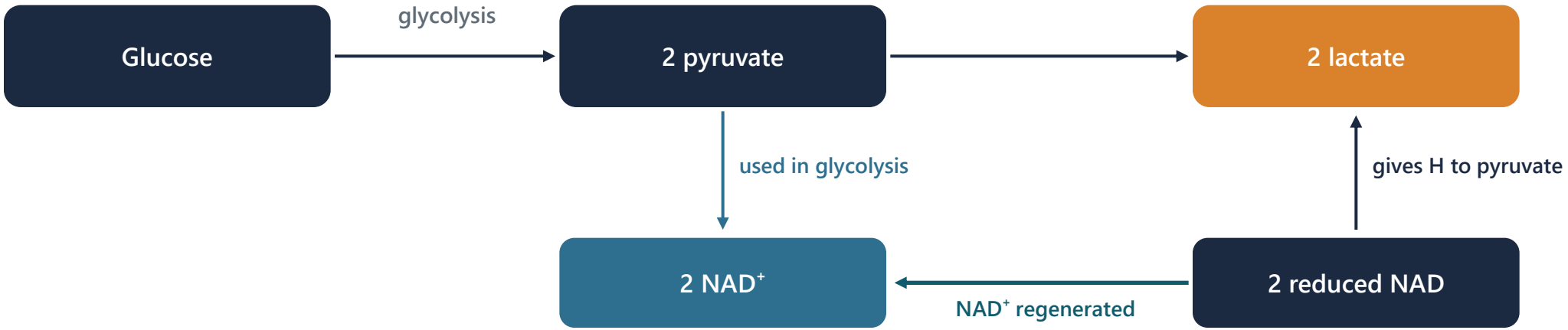
In the cytoplasm, no oxygen needed.

+2 ATP (net)
+2 reduced NAD
2 pyruvate (3C)

Ten enzymes in all, one per step: you only need these four types of step.

No oxygen: recycling NAD with lactate

C1.2.9



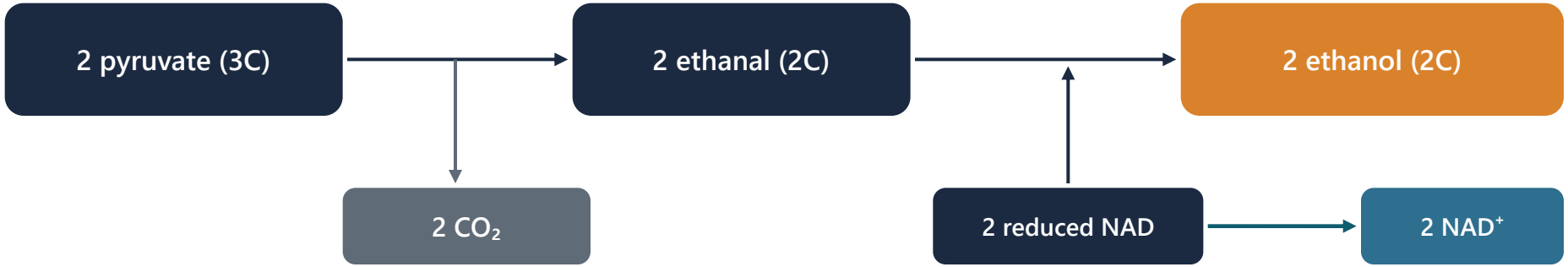
Glycolysis needs NAD^+. Without oxygen, reduced NAD cannot unload at the electron transport chain, so NAD^+ would run out and glycolysis would stop.

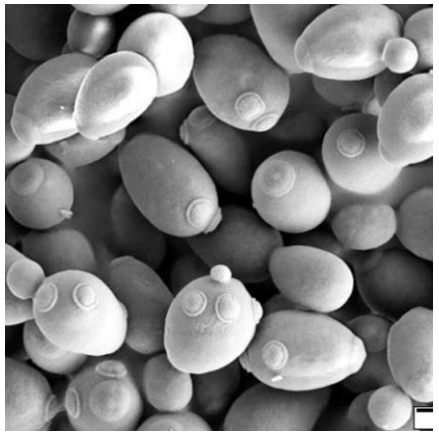
In humans, **pyruvate** takes the hydrogen and becomes **lactate**. This frees NAD^+, so glycolysis keeps making its **2 ATP**. No extra ATP is made.

Remember

Making lactate is about regenerating NAD^+, not making ATP. Later, lactate is taken to the liver and converted back, using oxygen.

Yeast: ethanol and carbon dioxide





Yeast (SEM). M. D. Murtey and P. Ramasamy, CC BY 3.0.

	Humans	Yeast
Glycolysis	Same	Same
Hydrogen acceptor	Pyruvate	Ethanal
Products	Lactate	Ethanol + carbon dioxide

Same purpose in both: regenerate NAD^+ so glycolysis continues. Only the hydrogen acceptor and products differ.

Using yeast: baking and brewing



Dough rising. Wald1siedel, CC BY-SA 4.0.



Beer fermenting. Picardin, CC BY 4.0.



Yoghurt. Julikalucky, CC0.

Baking

Carbon dioxide makes the dough rise. Ethanol evaporates in the oven.

Brewing

Ethanol is the product. Fermentation stops at about 12–15% ethanol, which kills the yeast.

Yoghurt

Bacteria make lactic acid (like our muscles). The acid thickens and preserves the milk.

Fast but wasteful

C1.2.9



200 m final, Osaka 2007. Eckhard Pecher, CC BY 2.5.

Per glucose	Anaerobic	Aerobic
ATP made	2	about 30–32
Needs oxygen?	No	Yes
Speed	Very fast	Slower

In a sprint, muscles use ATP faster than blood can supply oxygen, so they also respire anaerobically.

Most of the energy of glucose is left in lactate or ethanol: that is why ethanol burns as a fuel. Fast, but only a short-term fix.

Working with data: yeast and sugar

C1.2.10

Carbon dioxide released by yeast in 20 minutes at 30 °C, with different glucose concentrations. These data are invented for practice; the pattern is realistic.

Glucose / %	CO_2 / cm^3
0	0.2
2	4.1
4	7.6
8	10.8
16	11.2
32	6.5

1. Describe the effect of glucose concentration on the volume of CO_2.
2. Explain the increase up to 8%.
3. Suggest why the volume levels off between 8 and 16%.
4. Suggest why it falls at 32%.

Check-in: class 1

C1.2.7–10

1. Explain why removing hydrogen is oxidation.
2. State how many ATP (after costs) and reduced NAD glycolysis gives per glucose.
3. Name the four types of step in glycolysis.
4. Explain why pyruvate is turned into lactate when oxygen is short.
5. Compare the products of anaerobic respiration in humans and yeast.
6. State which yeast product matters in baking and which in brewing.

How to use this

Answer without your notes, then check against slides 4–10.

Into the mitochondrion

C1.2.11

Glycolysis
cytoplasm

→

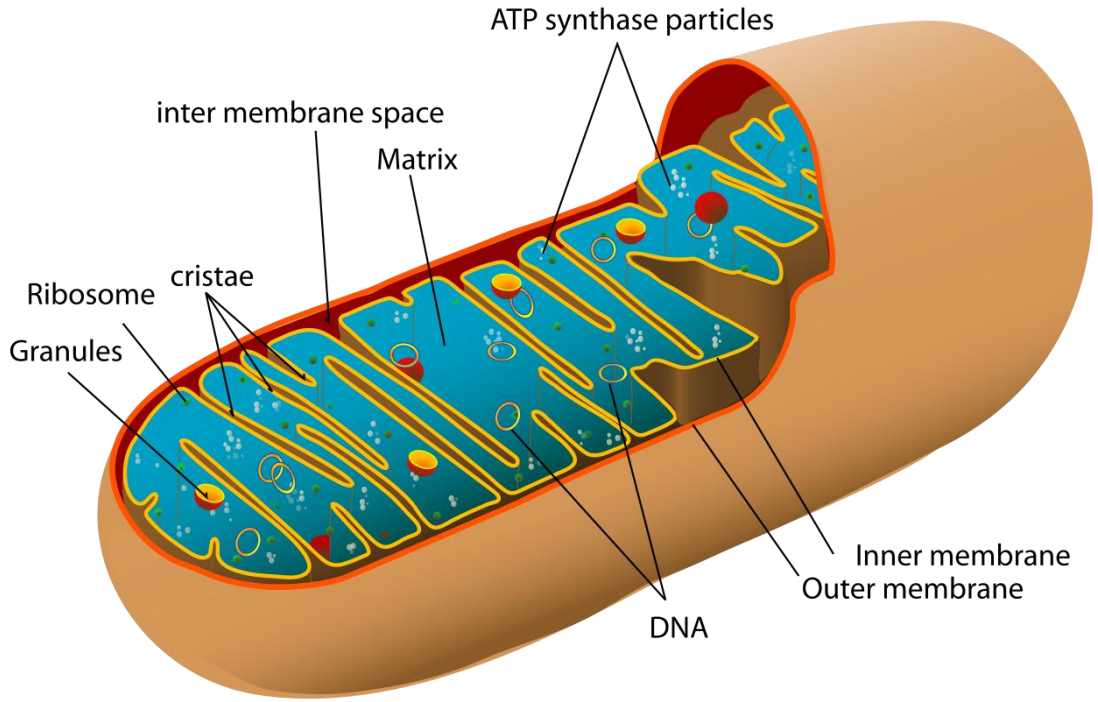
Link reaction
matrix

→

Krebs cycle
matrix

→

Electron transport chain and chemiosmosis
inner membrane



Mitochondrion. Mariana Ruiz Villarreal, public domain.

With oxygen, pyruvate is carried into the mitochondrion.

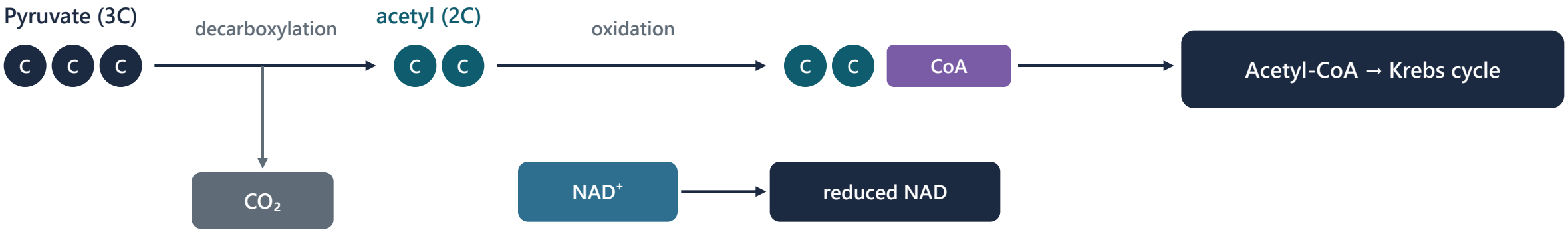
Matrix: link reaction and Krebs cycle.

Inner membrane (folded into **cristae**): electron transport chain and ATP synthase.

Intermembrane space: the narrow gap where protons build up.

The link reaction

C1.2.11



Pyruvate loses one carbon as **CO_2** (**decarboxylation**) and loses hydrogen to NAD^+ (**oxidation**).

The 2C **acetyl group** left over is carried into the Krebs cycle by **coenzyme A**.

Fats enter here too: fatty acids are cut into acetyl groups that join coenzyme A directly.

Per glucose

2 pyruvate →
2 acetyl-CoA
2 CO_2
2 reduced NAD
0 ATP

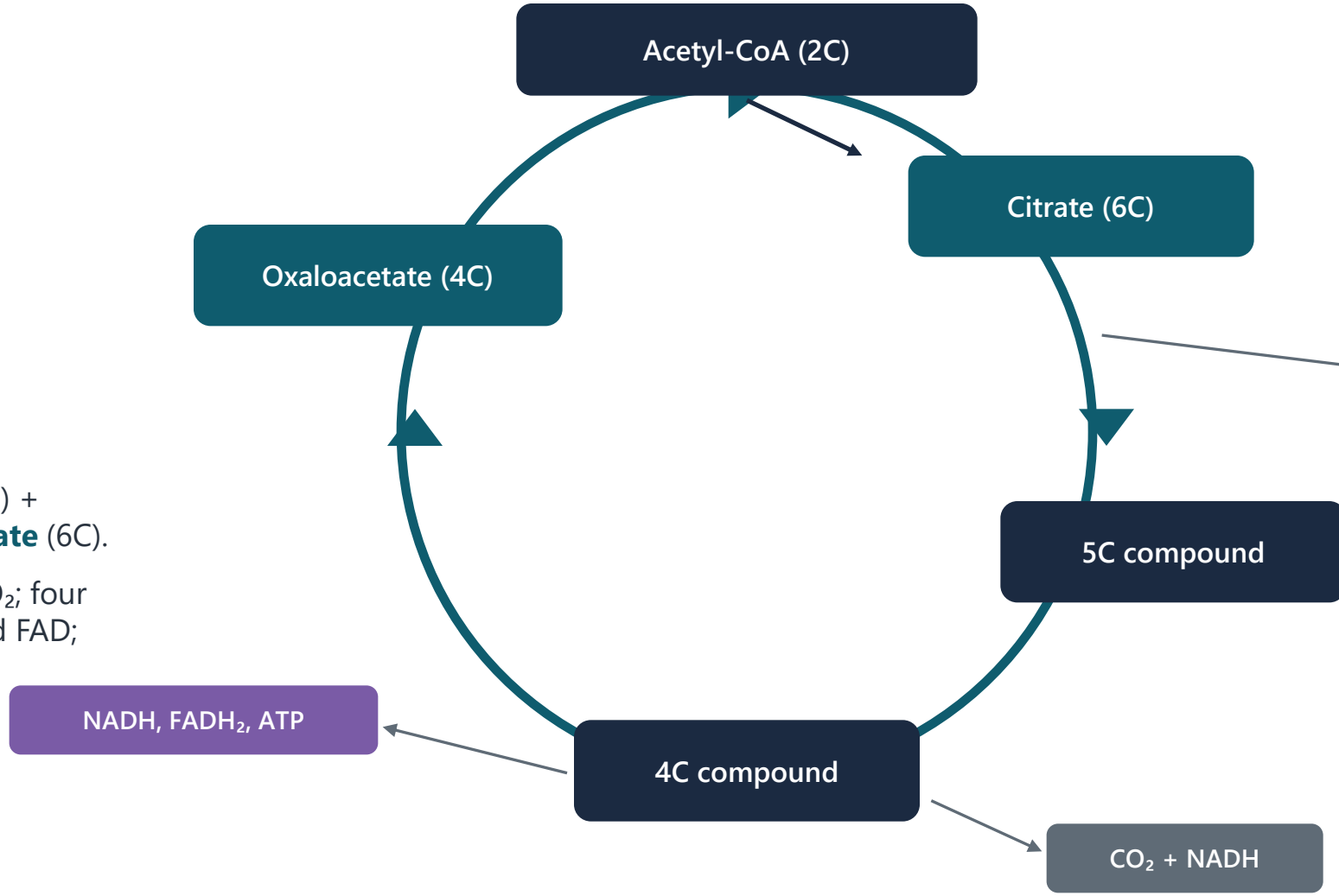
The Krebs cycle



Hans Krebs. Public domain.

In the **matrix**. Acetyl (2C) + oxaloacetate (4C) → **citrate** (6C).

Two carbons leave as CO_2; four oxidations load NAD and FAD; oxaloacetate is rebuilt.



Per turn	
CO_2	2
NADH	3
$FADH_2$	1
ATP	1

Where did the glucose go?

C1.2.8–12

Per glucose	CO_2	ATP	Reduced NAD	Reduced FAD
Glycolysis	0	2 (net)	2	0
Link reaction	2	0	2	0
Krebs cycle (2 turns)	4	2	6	2
Total so far	6	4	10	2

The carbon

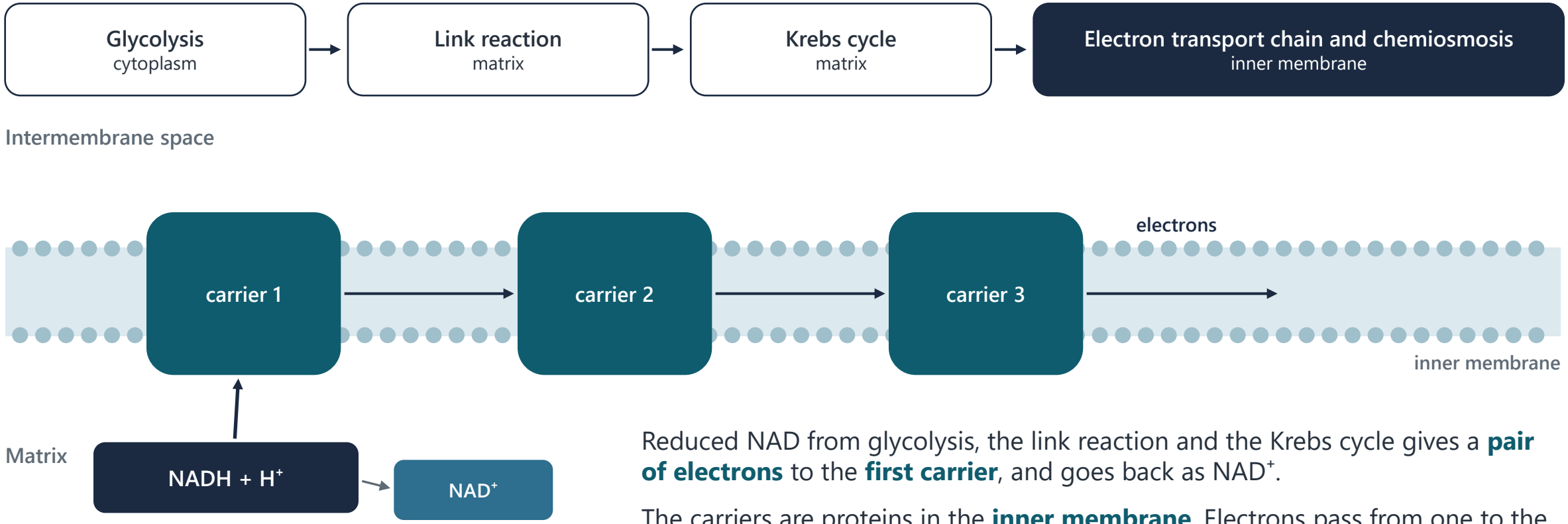
All 6 carbons of glucose have left as CO_2: this is the CO_2 you breathe out.

The energy

Only 4 ATP so far. The rest is held in 10 reduced NAD and 2 reduced FAD, ready for the electron transport chain.

The electron transport chain

C1.2.13

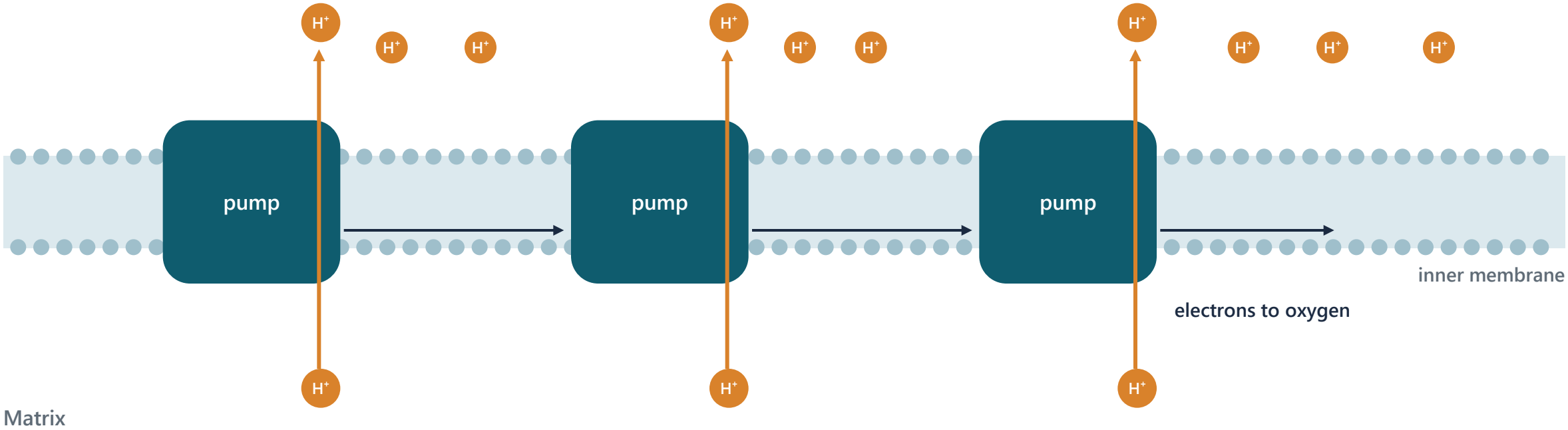


Reduced NAD from glycolysis, the link reaction and the Krebs cycle gives a **pair of electrons** to the **first carrier**, and goes back as NAD^+.

The carriers are proteins in the **inner membrane**. Electrons pass from one to the next, releasing energy in small steps.

The energy pumps protons

C1.2.14

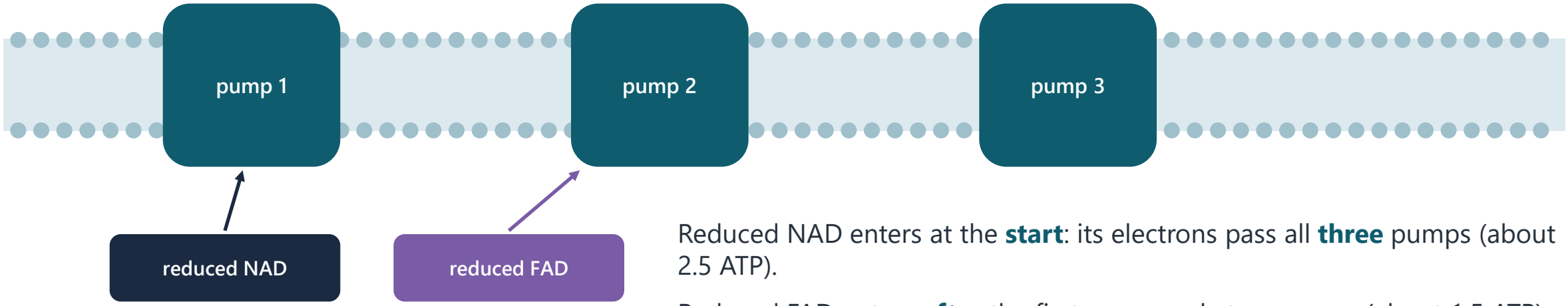


As electrons pass through, three carriers use the released energy to pump **protons (H^+)** out of the matrix into the **intermembrane space**.

The membrane does not let protons leak back, so a **proton gradient** builds up: a store of energy, like water behind a dam.

Why FAD gives less ATP

C1.2.13–14



Reduced NAD enters at the **start**: its electrons pass all **three** pumps (about 2.5 ATP).

Reduced FAD enters **after** the first pump: only two pumps (about 1.5 ATP).

In short

Same electrons, fewer pumps, fewer protons, less ATP.

Working with data: blocking the Krebs cycle

C1.2.12

Mitochondria were given pyruvate. A poison that blocks one Krebs cycle enzyme was added at 10 minutes. These data are invented for practice.

Time / min	Citrate	CO_2 released
0	10	100
5	10	100
10	10	100
15	45	30
20	80	10

1. Describe what happens to citrate and CO_2 after the poison is added.
2. Suggest which part of the cycle is blocked. Explain.
3. Predict what happens to oxygen uptake. Explain.

Check-in: class 2

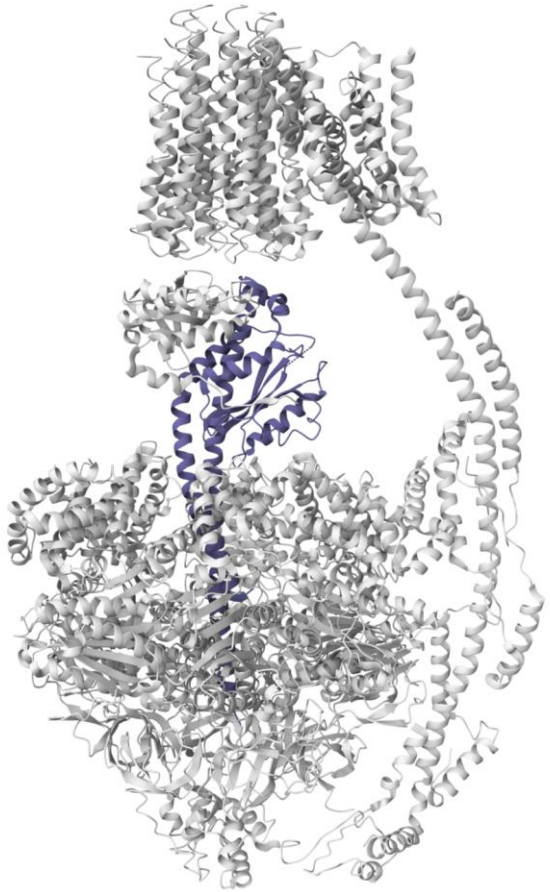
C1.2.11–14

1. Name the two changes to pyruvate in the link reaction.
2. Name the 4C and 6C compounds in the Krebs cycle.
3. State the products of one turn of the Krebs cycle.
4. Describe what reduced NAD does at the electron transport chain.
5. State the direction in which protons are pumped.
6. Explain why reduced FAD gives less ATP than reduced NAD.

How to use this

Draw the Krebs cycle from memory first, then answer.

ATP synthase: a tiny motor



ATP synthase; the rotating stalk is highlighted. Zhou et al., CC0.

Protons flow down the gradient through the channel

↓

The rotor and stalk spin

↓

The head changes shape

↓

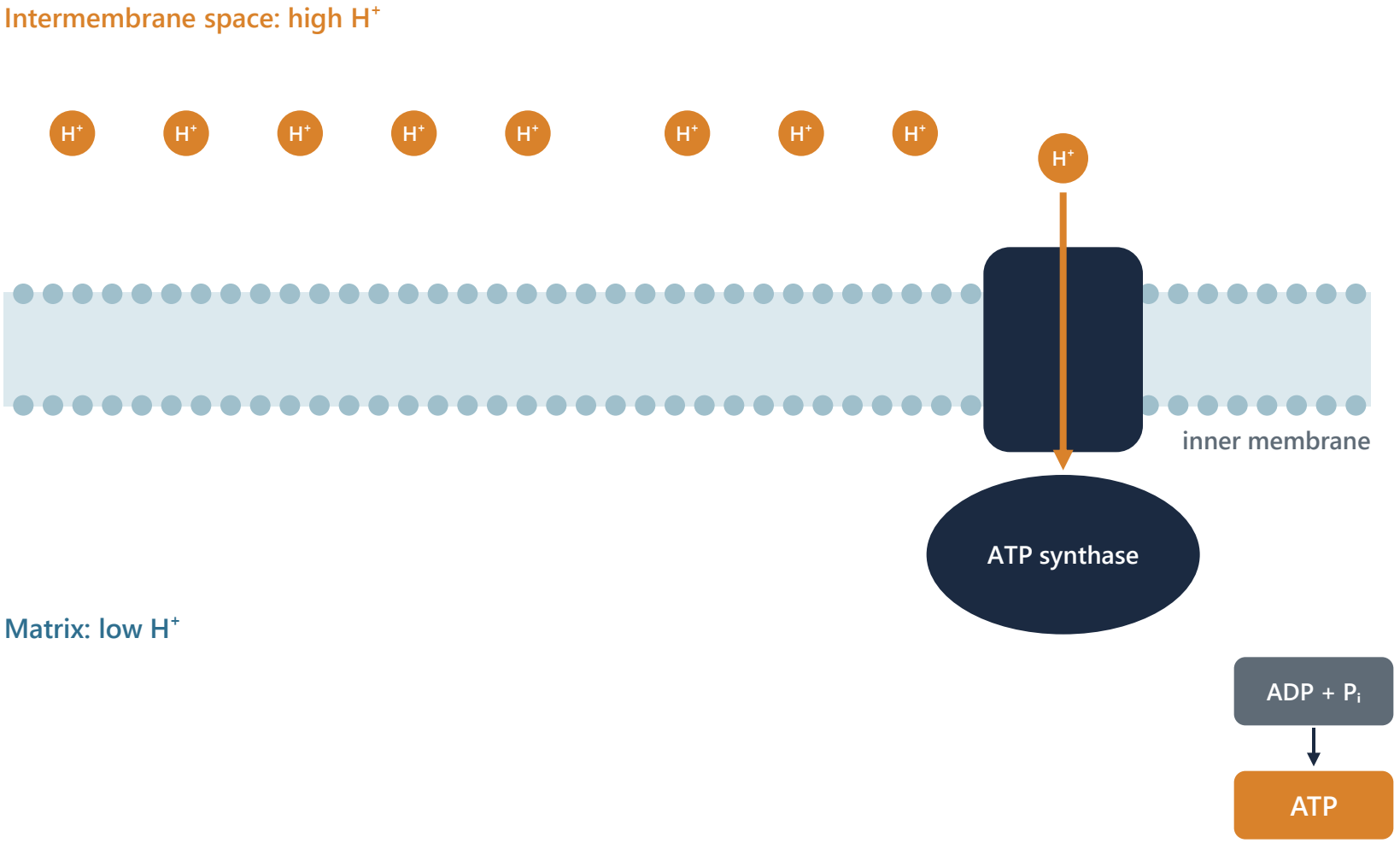
$ADP + P_i \rightarrow ATP$

Protons are not used up: their movement supplies the energy.

Coupling

ATP synthase links the energy of protons flowing back to the joining of phosphate to ADP. No flow, no spin, no ATP.

Chemiosmosis



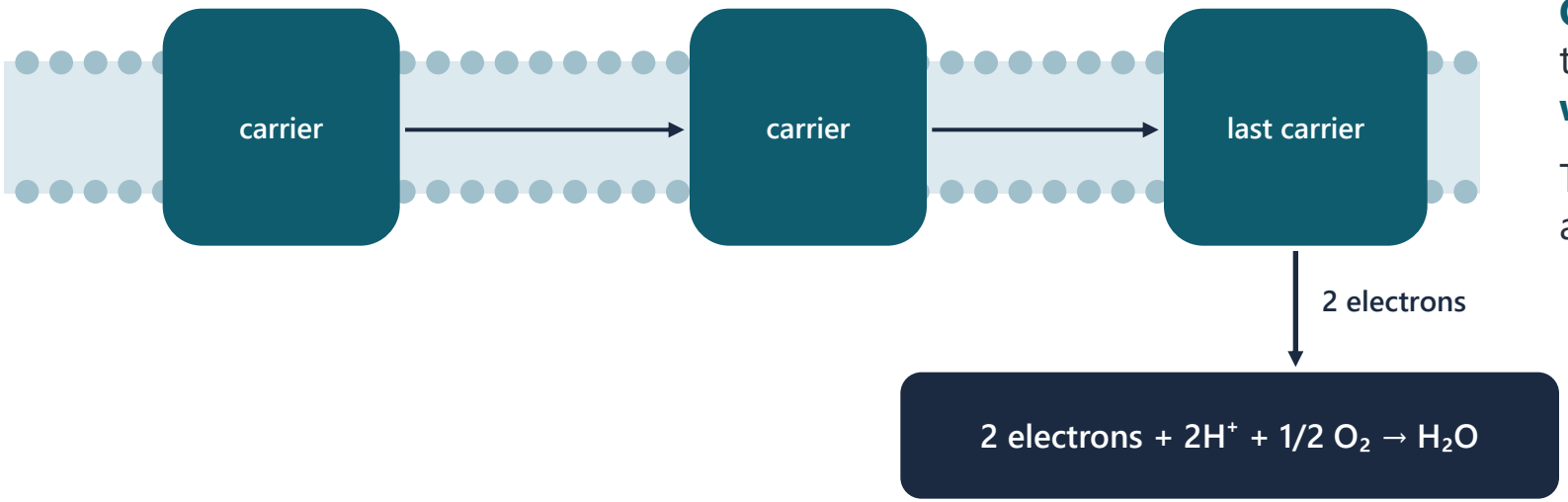
Protons can only get back into the matrix through **ATP synthase**.

As they flow **down their gradient**, ATP synthase uses the energy to make **ATP**.

Making ATP with a proton gradient across a membrane is called **chemiosmosis**.

Oxygen: the final electron acceptor

C1.2.16



Oxygen takes the electrons at the end of the chain and joins with protons, making **water**.

This keeps electrons flowing, so pumping and ATP production continue.

No oxygen?

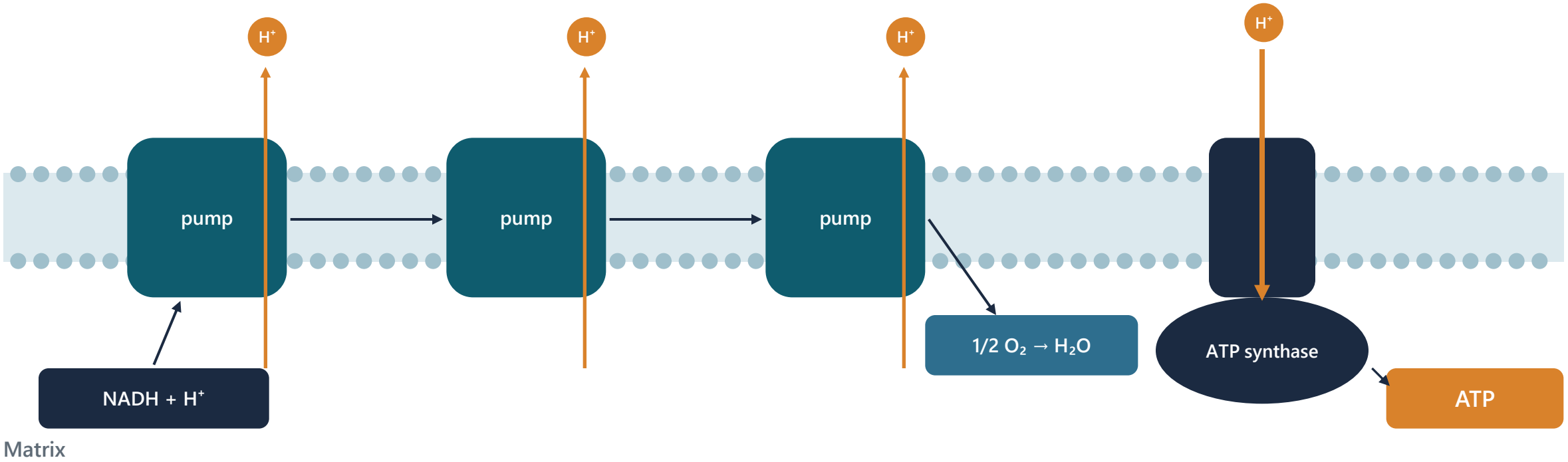
Electrons back up, pumping stops, reduced NAD cannot unload, so the Krebs cycle and link reaction stop too. Only glycolysis (with lactate) is left.

Cyanide

Blocks the last carrier, so electrons cannot reach oxygen. ATP production collapses even though oxygen is present.

Putting it together

C1.2.13–16



Electron transport: electrons flow to oxygen and their energy pumps protons out. **Chemiosmosis**: protons flow back through ATP synthase and their energy makes ATP.

Together: **oxidative phosphorylation**, which makes most of the ATP.

Counting the ATP

C1.2.8–16

Per glucose	Made directly	Carriers	ATP from carriers
Glycolysis	2 (net)	2 NADH	5
Link reaction	0	2 NADH	5
Krebs cycle	2	6 NADH, 2 $FADH_2$	15 + 3
Total	4		28: about 32 in all

Why "about"?

Using 2.5 ATP per NADH and 1.5 per $FADH_2$. Some energy is spent moving molecules across membranes, so real yields are nearer 30. Older books say 36–38 (using 3 and 2).

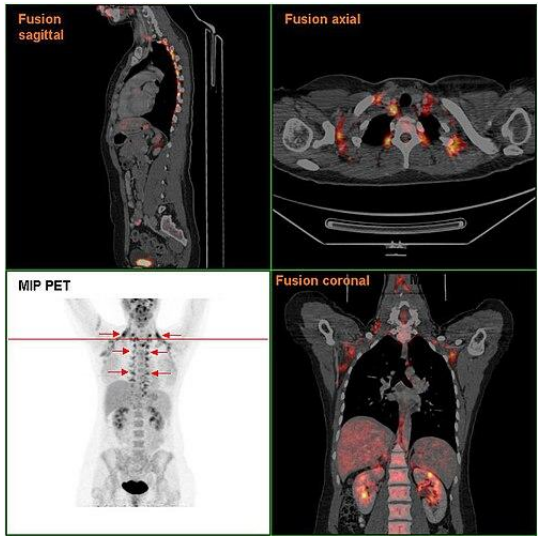
Working with data: poisons and brown fat

C1.2.15–16

Oxygen use and ATP made by mitochondria given pyruvate (% of control). These data are invented for practice; the pattern matches classic experiments.

Condition	Oxygen use	ATP made
Control	100	100
Cyanide	3	2
DNP (lets H^+ leak through the membrane)	160	5

1. Explain the cyanide results.
2. Explain why DNP raises oxygen use but stops ATP production.
3. Suggest where the energy goes with DNP.



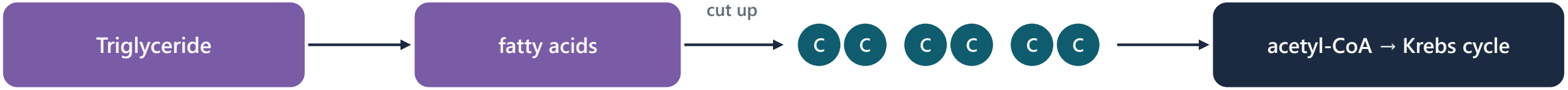
Brown fat (glowing) on a PET-CT scan. Hg6996, public domain.

Brown fat

Has a protein that lets protons leak back on purpose, so the energy becomes heat. It keeps babies and hibernating mammals warm.

Fats as fuel

C1.2.17



	Glucose	Fatty acid
Formula	$C_6H_{12}O_6$	$C_{16}H_{32}O_2$
Energy per gram	about 17 kJ	about 37 kJ

Fatty acids have **less oxygen** and **more hydrogen** to oxidize, so more reduced NAD and FAD, so **more ATP per gram**.

They skip glycolysis, so fat can only be respired **aerobically**.

So
Sprints run on carbohydrate; long, slow exercise uses more fat.

Fats and carbohydrates compared

	Carbohydrate	Lipid
Energy per gram	About 17 kJ	About 37 kJ
Route into respiration	Glycolysis	Acetyl-CoA
Anaerobic use?	Yes	No
Speed of release	Fast	Slower
Metabolic water	Less	Much more
Stored in cells	Soluble: osmotic effect	Insoluble: none



Bactrian camel, Gobi Desert. Bernard Gagnon, CC0.

Metabolic water

Respiring fat makes a lot of water. A camel's hump is fat, not a water tank.

The whole pathway on one slide

C1.2.7–17

Stage	Where	In	Out (per glucose)	Key idea
Glycolysis	Cytoplasm	Glucose, 2 ATP, 2 NAD^+	2 pyruvate, 4 ATP, 2 NADH	Phosphorylation, lysis, oxidation, ATP formation
Link reaction	Matrix	2 pyruvate	2 acetyl-CoA, 2 CO_2, 2 NADH	Decarboxylation and oxidation
Krebs cycle	Matrix	2 acetyl-CoA	4 CO_2, 6 NADH, 2 $FADH_2$, 2 ATP	4 oxidations, 2 decarboxylations
Electron transport and chemiosmosis	Inner membrane	10 NADH, 2 $FADH_2$, O_2	About 28 ATP, water	Proton gradient drives ATP synthase

Common mistakes to avoid

Not this	But this
Lactate formation makes extra ATP.	It makes none: it regenerates NAD^+.
Glycolysis happens in the mitochondrion.	In the cytoplasm; link reaction and Krebs cycle in the matrix.
Oxygen is used in the Krebs cycle.	Only at the end of the electron transport chain.
The electron transport chain makes ATP.	It builds the proton gradient; ATP synthase makes ATP.
Protons are pumped into the matrix.	Out of the matrix, into the intermembrane space.

Check that you can...

1. **Explain** oxidation as removal of hydrogen, and NAD as a hydrogen carrier
2. **Outline** glycolysis and its net yield
3. **Explain** how lactate and ethanol formation regenerate NAD, and uses of yeast
4. **Explain** the link reaction and the Krebs cycle
5. **Explain** how the electron transport chain builds a proton gradient
6. **Explain** chemiosmosis and the role of ATP synthase
7. **Explain** why oxygen is the final electron acceptor
8. **Compare** fats and carbohydrates as fuels for respiration

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State how many ATP (after costs) and reduced NAD glycolysis gives per glucose.
2. Explain why pyruvate is converted to lactate when oxygen is short.
3. Explain how the flow of protons through ATP synthase produces ATP.

Image credits

Animal mitochondrion diagram en.svg – Mariana Ruiz Villarreal LadyofHats – Public domain

Mitochondria, mammalian lung – TEM.jpg – Louisa Howard – Public domain

Saccharomyces cerevisiae SEM.jpg – Mogana Das Murtey and Patchamuthu Ramasamy – CC BY 3.0

Bread dough in baskets 02.JPG – Wald1siedel – CC BY-SA 4.0

Anchor Brewing Company fermentation tank.JPG – Picardin – CC BY 4.0

Yoghurt and spoon.jpg – Julikalucky – CC0

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ATP5C within Bos taurus mitochondrial ATP synthase state 1a 5ARA.png – Zhou A, Rohou A, Schep DG, Bason JV, Montgomery MG – CC0

Brownfat PETCT.jpg – Hg6996 – Public domain

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